

ENME 435 – Remote Sensing: Spring 2026
Department of Mechanical Engineering**Due Date** Friday, April 24th, 2026 by 11:59 pm**Submission Information**

- Submit .pdf response to Question #3 via Gradescope

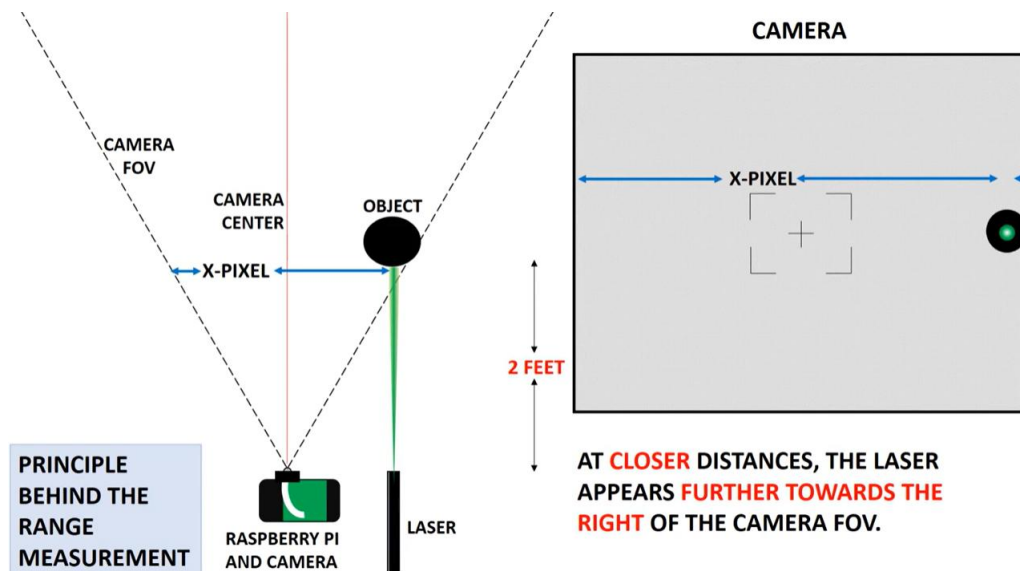
Alignment of field deployable lidar, Inertial Measurement Unit (IMU), & planning for semester video

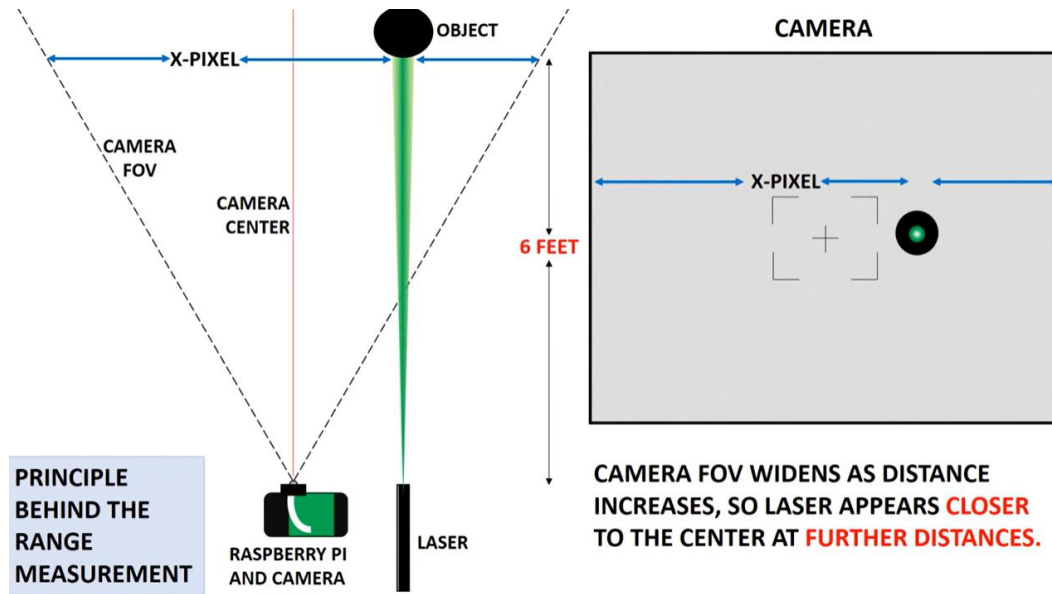
Reminder: keep recording pictures & videos of each aspect of your project to edit/stitch together in your semester project video!

Reading: In preparation for our guest lecture from Dr. Phil Sandborn, please read through the “Understanding the Physics of Coherent Lidar” pdf available on ELMS > Assignments > HW #6.

Question #1 (nothing to submit)

In Assignment #5, and in lecture, we explored the theory behind our geometrical range measurement using a single beam laser and a camera. By calibrating the bistatic setup, we are able to collect range measurements solely by tracking the x- coordinate of the center of the imaged laser beam:





Be sure to review:

https://www.youtube.com/watch?v=xOO9d_6w2PE&list=PLekqOMAzgrTRP6Ms7wI37-2IKjU92tqnL&index=24

Here we expand upon the single beam technique, replacing the single beam laser with a line laser, which *dramatically* improves the data acquisition rate of the lidar.



Single beam laser bistatic configuration (left, prior semesters) vs. line laser bistatic configuration (right, current semester)

This sensor configuration requires angular alignment of the line laser with the Raspberry Pi camera. To begin, head over to GitHub and download *laser_alignment_image.py*:

<https://github.com/oneshell/enme489y>

This script opens the RPi camera feed to support the alignment process (**note**: you may have to edit the script based on your setup!). The *laser_alignment_image.py* script uses OpenCV to plot a series of

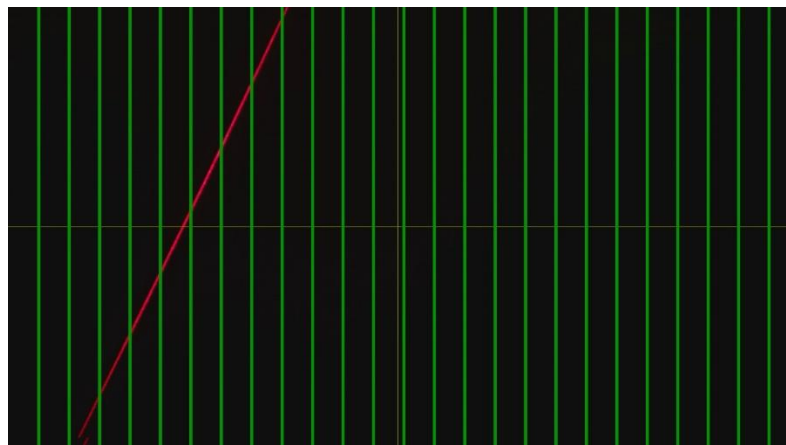
vertical green lines along the camera frame. The spacing of these lines can be adjusted as desired. Begin the alignment process by connecting the LED line laser to a 5V power source, such as the Arduino Uno.

*Note: you will have to **modify** the provided scripts for this part of the assignment. This is intentional, as part of the learning exercise.*

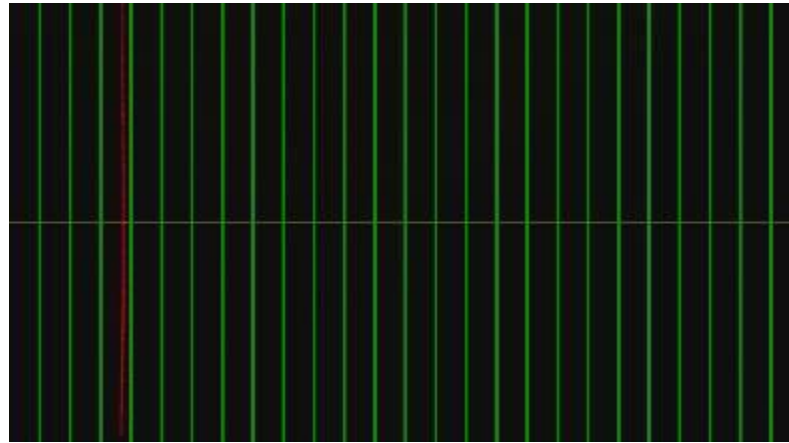
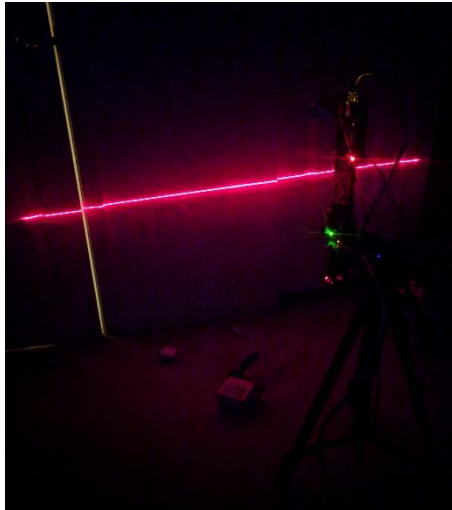
Regardless of how you have mechanically integrated your sensor, **it is imperative that the RPi camera be rigidly mounted prior to beginning the alignment process**. Once the lidar is aligned, the axis of the camera and the axis of the laser should remain aligned and parallel throughout the semester, so ensure the RPi camera is securely mounted before continuing!

While monitoring the camera's video feed, rotate and align the line laser such that the red laser line is parallel to the green alignment lines (see example images below). Leave the video stream running and **secure the laser in place** (e.g. using glue/epoxy, set screw, etc.).

After properly securing the laser, ensure the alignment of the laser line has not shifted. Double check that both the RPi camera and line laser are rigidly secured in their respective mounts. Pressing the **q** key will end the video feed.



Line laser orientation at the beginning of the alignment process (not currently aligned). Note the lack of parallelism between the red laser line and the green vertical alignment lines.



Line laser aligned. Red laser line now parallel with green vertical alignment lines.

Again: it is **imperative** that the RPi camera and laser be **rigidly mounted** to maintain their alignment and ensure proper functioning of the lidar sensor throughout the semester.

Compared to the theoretical single laser beam configuration of Assignment #5, integration of the line laser is equivalent to stacking multiple single beam LED lasers on top of each other to form an array of beams or, in this case, a line.

In effect, each row of pixels of the camera frame is now a single laser beam lidar!

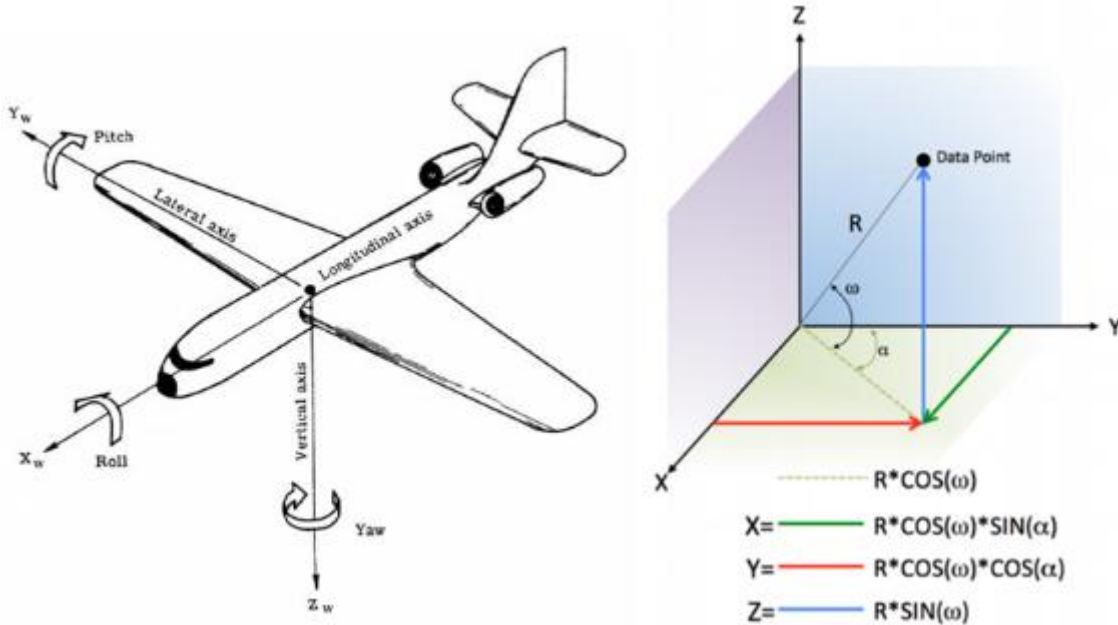
Thought: consider **recording videos** of the laser alignment process, both with your cell phone *and* with your RPi, to use in your semester video!

Question #2 (nothing to submit)

As discussed in lecture, to geolocate lidar data requires the sensor to record:

1. Range data, describing the distance between the sensor and the target
2. GPS data, describing the latitude-longitude-altitude position of the sensor
3. IMU data, describing the pitch-roll-yaw pointing of the sensor

With these sources of information, each lidar range measurement can be transformed into a XYZ coordinate in 3-dimensional space.

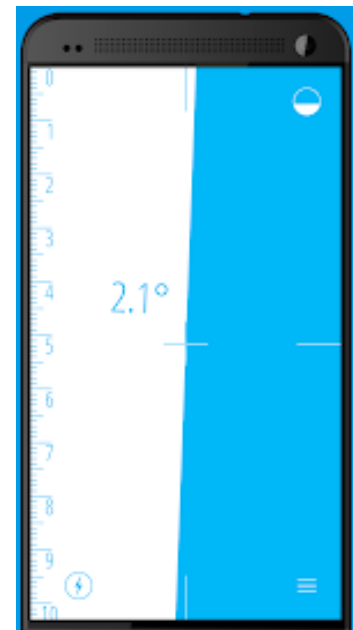


For ENME435, we collected GPS data in Assignment #1 to identify the location of each target on campus.
No further GPS data is required.

Now that we have the line laser installed and aligned, let's evaluate how to record IMU data with our lidar sensor. To measure the pointing angle of the RPi lidar, use the accelerometer in your cell phone via an app such as Bubble Level (for Android) or equivalent.

To begin, **securely** mount the phone to the lidar mounting plate. Enter the angle reading provided by the app when saving range data in Python (more information on this coming in Assignment #7).

With the bistatic lidar and an IMU (here the cell phone equivalent), each lidar range measurement can be transformed into a XYZ coordinate!



Question #3 (40 points)

Here we utilize the triangulation lidar range measurement from Assignment #5 and upgrade the sensor calibration from a single beam laser to the line laser.

Identify a wall free of obstructions and setup the lidar orthogonal to the wall surface (image at right).



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Head over to GitHub and download *laser_alignment_image_blank.py*:

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<https://github.com/oneshell/enme489y>

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This script opens the RPi camera feed to support the alignment process (note: you may have to edit the script based on your setup). The

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laser_alignment_image_blank.py script asks the user to “**Please enter distance from wall, in inches:** “. Enter the distance measured with a tape measure and hit the Enter key. The script will echo the distance value you entered into the terminal window (to confirm the number) then opens the video feed.

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*Note: you will have to **modify** the provided scripts for this part of the assignment. This is intentional, as part of the learning exercise.*

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Notice that the distance number has now been added to the top right corner of the video feed via the **cv2.putText()** command.

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With the instrument aligned orthogonal to the wall, hit the **m** key. This will save a .jpg file on your RPi, the filename of which is defined by the distance from the wall. To confirm you have successfully recorded the .jpg, in a terminal window type:

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ls

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xdg-open distance_value_here.jpg

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This will open the saved .jpg file.

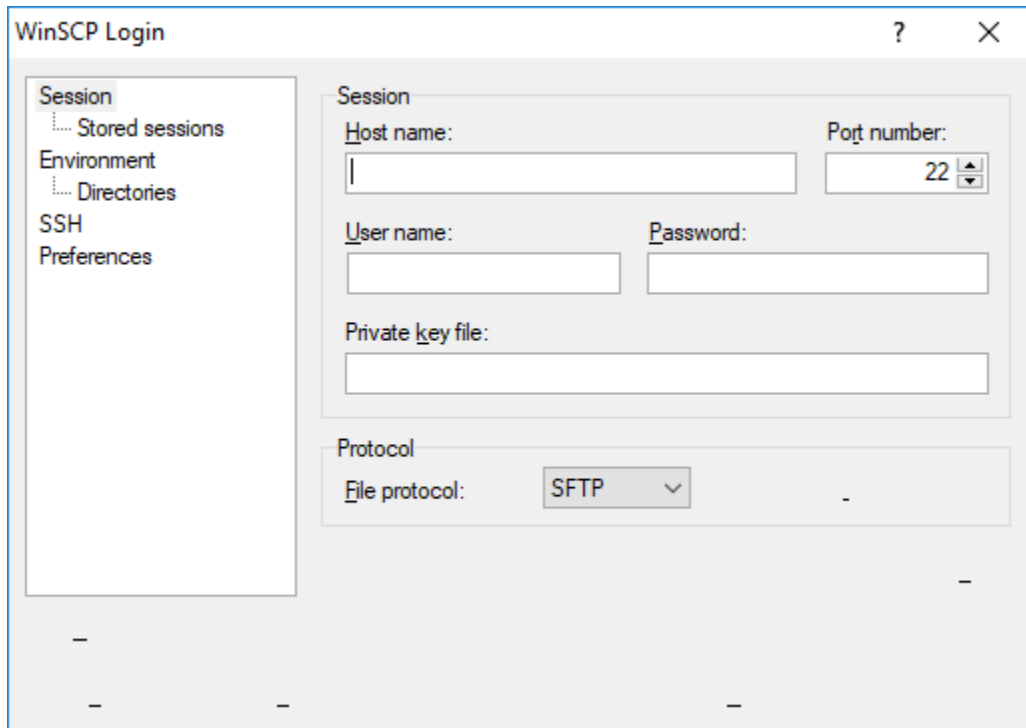
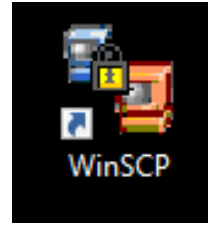
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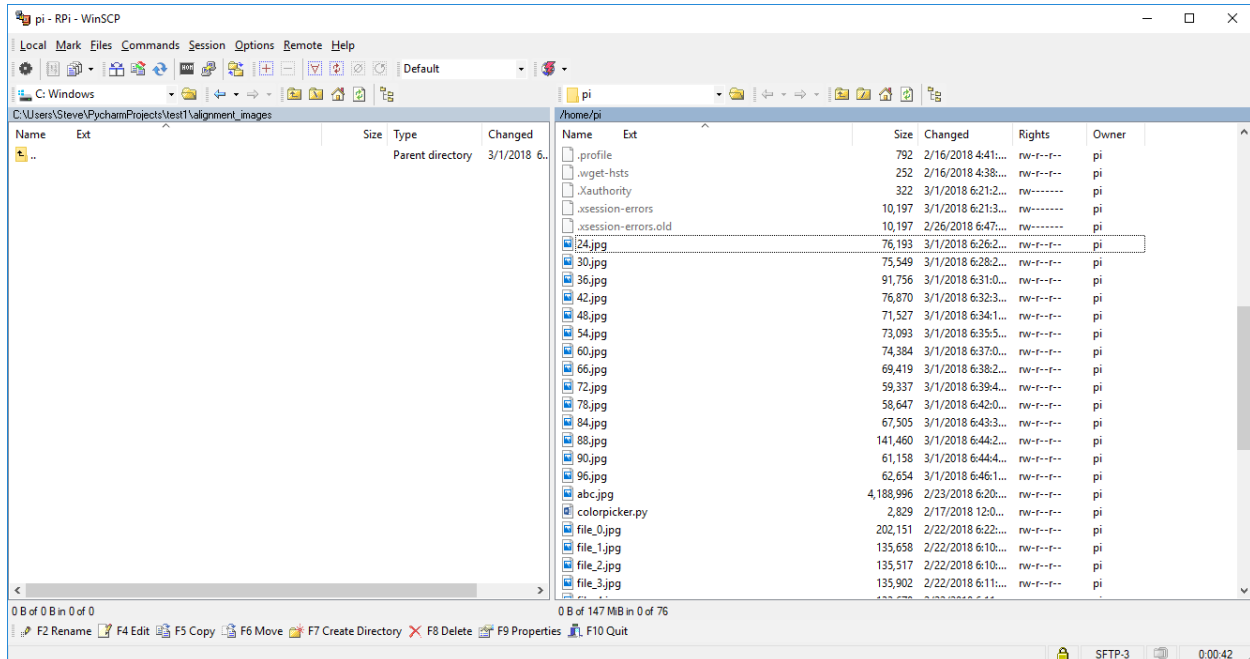
Run the *laser_alignment_image_blank.py* script and save .jpg files while ranging to the wall, for known distances from 0 to 10 feet, in increments of 6 inches. An example of this sequence is shown in the collage at left. Once complete, type **ls** into a terminal window and confirm the .jpg files have successfully saved to your RPi.

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Transfer all of the .jpg files created during this exercise from your RPi to your laptop. This can be accomplished in several ways:

1. Upload images to Gmail and email files
2. Insert a USB into your RPi and copy files
3. My favorite: **FTP into the RPi** from your laptop using an FTP client such as WinSCP. The Host name will be the IP address of your RPi. The User name and Password are the same as when using Putty and VNC Viewer (typically **pi** and **raspberr** or **raspberry**).





With all .jpg files transferred to your laptop, specifically your working directory in Anaconda (or PyCharm, etc.), head over to GitHub and download *laser_alignment.py*:

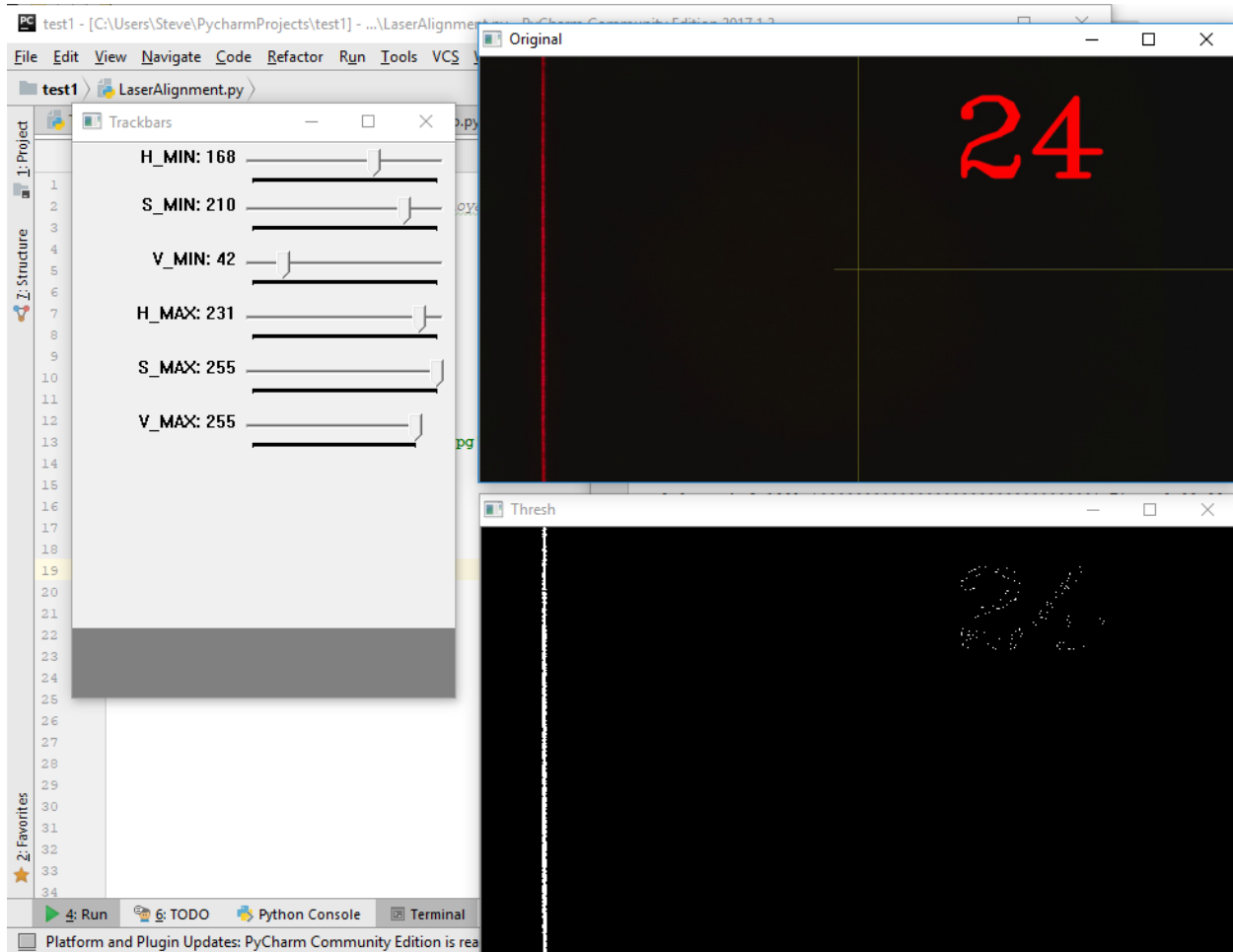
<https://github.com/oneshell/enme489y>

Let's explore this script, which will be executed in Anaconda (or PyCharm, etc.). When writing this assignment, Dr. Mitchell placed his alignment images in a folder on his laptop named **alignment_images**. (**Important:** avoid numbers in folder names!) Therefore, the *laser_alignment.py* script initially finds all .jpg files located in the /alignment_images folder.

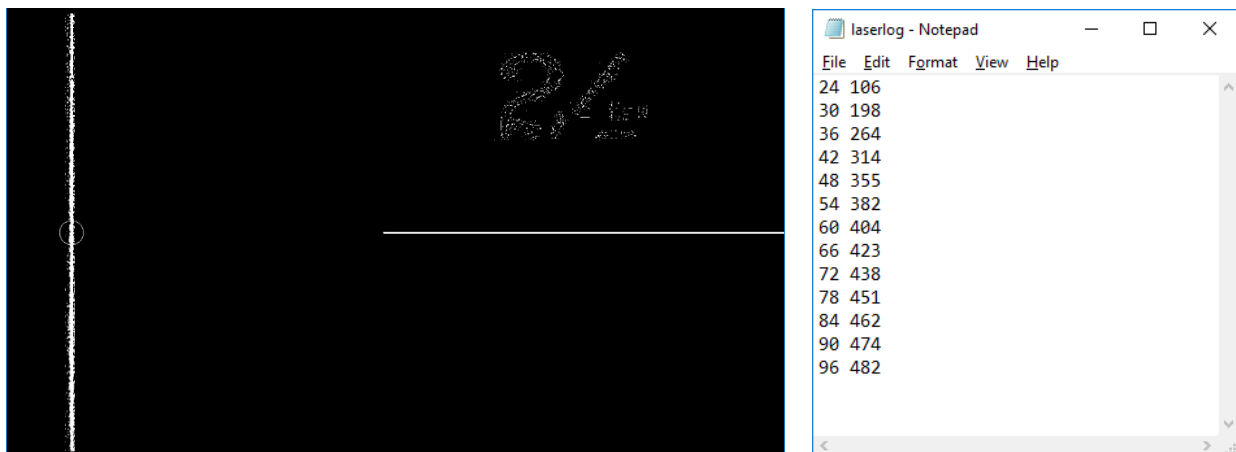
Consider updating the code as required for your setup:

```
files = glob.glob('alignment_images/*.jpg')
```

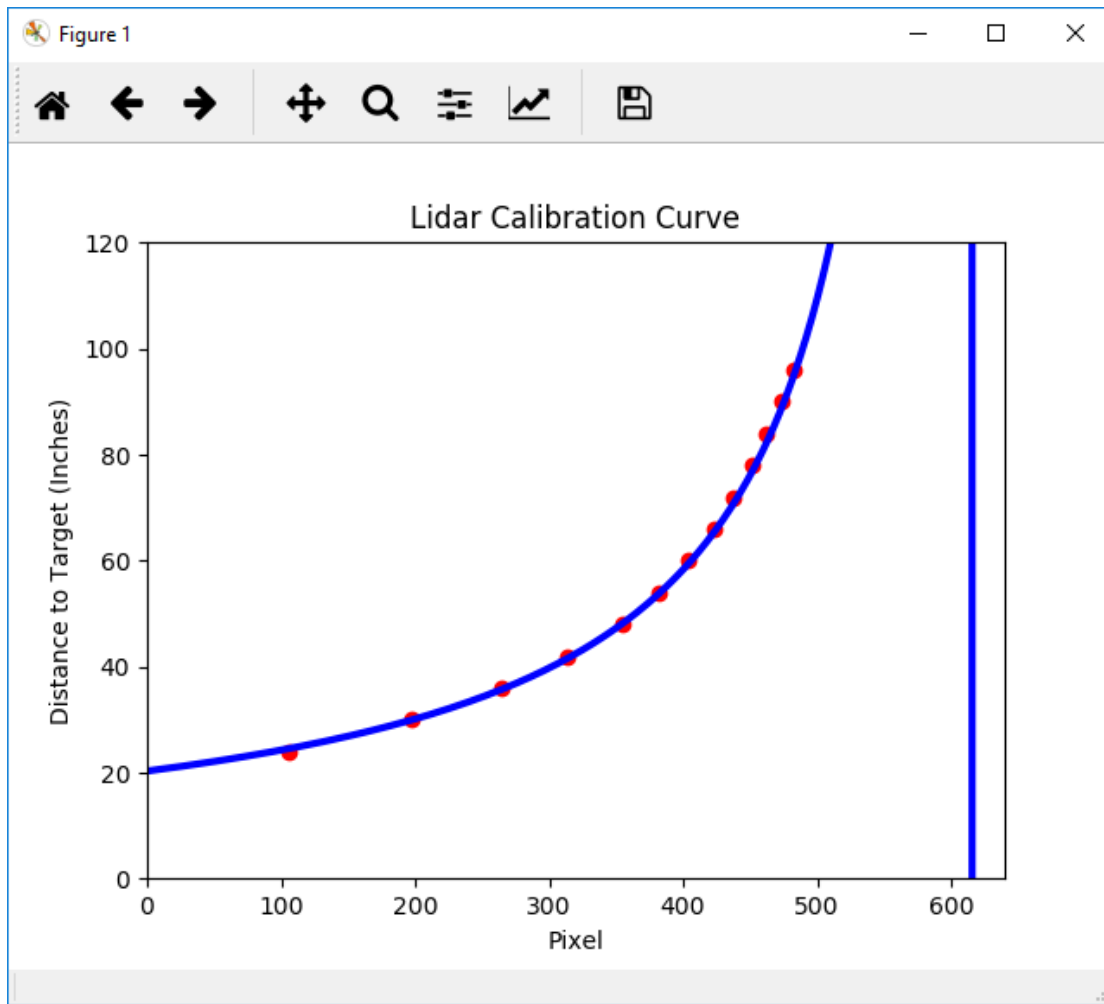
The script then defines the lower and upper color bounds to apply when masking each image (just like we did with our lane detection algorithm!). These can be tuned using the *colorpicker.py* script:



Execute the `laser_alignment.py` script in Anaconda (or PyCharm, etc.). The script loops through each .jpg file in the directory, identifies the red laser pixels in the center row of the image ($y = 360$ pixels), averages the pixels together to identify the center pixel, and then creates a `laserlog.txt` file which stores the distance and corresponding pixel coordinates to file.



Similar to the calibration theory of Assignment #5, the script then plots the acquired range data (red) on top of the theoretical output (blue). **Adjust the values of ro and rpc as required to calibrate your lidar.**



To complete this section of the assignment, ***paste your combined plot into a .doc file, generate 1-2 paragraphs describing how well your data aligns with the theoretical values, and upload as a .pdf file to Gradescope.***

Congratulations! Your field deployable lidar system is aligned, calibrated, and ready to head out on campus to map your target.

In Assignment #7 we will explore how to create 3D point clouds with the acquired range and IMU data, but plan to begin mapping your target *asap!*

Remember to take pictures & videos of yourself as you are on campus mapping for your semester project video and final presentation!

Question #4 (nothing to submit)

As detailed in the project writeup (ELMS > Modules > Project), the main deliverable of your project this semester is a YouTube video chronicling your project. Take a moment to reread the project requirements, emailing Dr. Mitchell (mitchels@umd.edu) with any questions or concerns.

What does an A-level project video look like? ...head over to the course YouTube playlist for examples.

Exemplary past project videos include:

<https://www.youtube.com/watch?v=jcYjsH325mU&list=PLekqOMAzgrTRP6Ms7wI37-2IKjU92tqnL&index=59>

https://www.youtube.com/watch?v=xOO9d_6w2PE&list=PLekqOMAzgrTRP6Ms7wI37-2IKjU92tqnL&index=24

<https://www.youtube.com/watch?v=VpbpRa6wk8E&list=PLekqOMAzgrTRP6Ms7wI37-2IKjU92tqnL&index=57>

<https://www.youtube.com/watch?v=jcYjsH325mU&list=PLekqOMAzgrTRP6Ms7wI37-2IKjU92tqnL&index=59>